

MATERIALS

The word “materials” describes the kind of matter we use for making and building things. Every object is made of a material—a hard material or a soft material, a rough or a smooth, a multicolored or a plain gray one. Nature has come up with millions of different materials, and people have developed millions more. You might think that is more than enough materials, but researchers are continually discovering amazing new natural materials, and inventing incredible new synthetic materials.

NATURAL OR SYNTHETIC?

People have used natural materials—such as wool, leather, and rubber—for thousands of years. Today we also make materials using chemicals. These synthetic materials have unique properties and make us less reliant on precious natural materials, but they can be difficult to dispose of in an environmentally friendly way.



Natural leather
Leather is made from animal skin. People have worn leather since the Stone Age, and still do. Leather can be molded into shape, retains heat, is fairly waterproof, and resists tears.



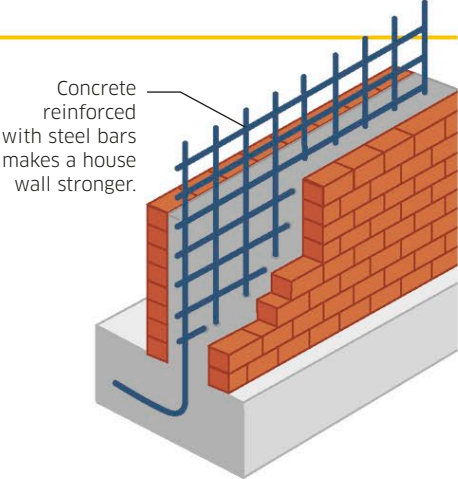
Synthetic trainer
The synthetic materials in this sports shoe offer several advantages over leather. They are easier and cheaper to produce, and have more flexibility, but they probably won't last as long.

CHOOSING MATERIALS

Different materials suit different purposes—there is no single “best material.” It all depends on what you want a material to do. Among their many properties, materials vary according to how hard they are, what they feel like, how strong or elastic they are, and whether or not they are waterproof.

Properties of materials
This chart lists some of the properties we need to think about when choosing a material for a certain product, and some common materials. Some of these properties are relative: marble, for example, is hard, but for a rock it is quite soft, which is why sculptors have chosen it to carve into statues since ancient times.

Composite materials
Sometimes the properties of one material are not enough, so two or more materials are combined into a composite. There are different composites. Concrete is made from strong stones, sand to fill the gaps, and cement to bind it all together. It stays together thanks to a chemical reaction that sets it. It can be made even stronger by adding steel bars in wet concrete. Fiberglass is a type of plastic reinforced with glass fibers. It is lightweight and easy to mold, and is used to make anything from bathtubs to boats and surfboards.



Material	Hardness	Texture	Strength	Elasticity	Water resistance
 Wood	From soft (balsa) to hard (mahogany)	Rough unless polished	Strength varies	Can be elastic or rigid	Some woods are more waterproof
 Glass	Very hard (does not flex under pressure)	Smooth	Not very strong; shatters on impact	Not elastic	Waterproof
 Diamond	One of the hardest materials known	Smooth when cut	Strong	Not elastic	Waterproof
 Marble	Hard (but soft for a rock)	Smooth	Strong	Not elastic	Waterproof
 Wool	Soft natural fibers	Rough or smooth	Strong fibers	Elastic in wool yarn and clothing	Not waterproof
 Kevlar®	Hard synthetic fibers	Smooth	Strong	Elastic	Waterproof
 Nylon	Hard synthetic	Smooth	Strong	Elastic in tights; less so in rope	Waterproof
 Steel	Hard metal alloy	Smooth	Strong	Elastic, particularly in springs	Waterproof
 Copper	Soft metal	Smooth	A weak metal	Not elastic	Waterproof